The λ hypothesis is falsified.

What five winters of Delhi NCR data actually say, and where the project goes now.

The aerosol–radiation–boundary-layer feedback loop gain λ does not separate pollution episodes that lock in from those that ventilate out. Held-out AUC is **0.504** — chance. The claim is withdrawn, and no part of the submission should rest on it.

THIS IS THE GATE WORKING, NOT THE PROJECT FAILING

The validation gate existed precisely to find this out in week one rather than week six. It cost one day of data engineering. The alternative was five people building a presentation around a number that would have collapsed under the first question from an NCMRWF reviewer.

WHAT REPLACED IT

Lock-in is not diagnosable from the present state. It is determined by weather that has not happened yet at episode onset. Ventilation measured **before** onset has no skill (AUC 0.514); measured **after** onset it is skilful (AUC 0.736). That is a forecasting problem, which is exactly what PS 26082 asks for — so the project stays on-brief without the λ claim.

1 What was built and run

Everything below is from real data, executed end to end. Nothing is projected or assumed.

COMPONENT	WHAT WAS OBTAINED
Meteorology	ERA5 via the Open-Meteo archive, 9-point grid across the NCR box, hourly 2020-10-01 to 2025-03-31. 39,432 hours. Boundary layer height, surface and top-of-atmosphere shortwave, cloud, temperature, RH, wind, pressure, precipitation. No Copernicus key was needed, which removed the only hard external dependency.
Ground PM2.5	34 OpenAQ sensors inside the NCR box, reference-grade networks only (CPCB, DPCC, UPPCB, HSPCB, IMD, IITM), anchored by the continuously-operating US Embassy monitor. 29,953 hours matched to the meteorology.
Episodes	161 labelled from the PM2.5 trajectory alone: 39 locked in, 122 ventilated, 11 in the held-out Novembers. Labels never saw λ.
Pipeline	Nine scripts, from synthetic self-test through elasticity estimation, episode labelling, validation, diagnosis and the reformulation attempt. All committed on branch ps26082-coupled-forecast.

A data limitation that must be stated

OPENAQ HAS A HOLE IN THE INDIAN RECORD

The old CPCB sensor generation stops around November 2022; the new generation begins February 2025. Nothing in between except the US Embassy monitor. Consequently 64% of in-season hours rest on two or fewer stations, including both holdout Novembers. The domain median is effectively one instrument across that stretch. This weakens spatial representativeness and is one of the candidate explanations examined in section 3.

SEASON	HOURS	MEDIAN STATIONS	PM2.5 P95
2020-21	3,171	15	365
2021-22	3,209	21	322
2022-23	2,071	1	334
2023-24	2,603	1	359
2024-25	3,375	1	347

2 The result

2.1 The mechanism is unmistakably present in the raw data

Before anything failed, this succeeded. The median diurnal cycle over five in-season winters is exactly what the physics predicts:

```
hour IST     PM2.5 (ug/m3)     BLH (m)     clearness
---------------------------------------------------
   23             163             44           -
   06             152             52           -
   09             151            196         0.554
   12             104            863         0.660
   15              79           1054         0.586
   18             116             60           -
```

PM2.5 falls by half as the mixed layer deepens twenty-five fold, and recovers as it collapses. The aerosol–boundary-layer relationship is not in doubt. **What failed is the attempt to extract a loop GAIN from it.**

2.2 λ returns chance

PREDICTOR	AUC (TRAIN)	HIT / FALSE ALARM	VERDICT
λ	0.504	0.86 / 0.73	chance
negative ventilation coefficient	0.421	0.42 / 0.33	worse than chance
ventilation-failure index (step 8)	0.554	0.31 / 0.17	chance
seasonal-anomaly ventilation index	0.512	0.62 / 0.52	chance

A hit rate of 0.86 alongside a false-alarm rate of 0.73 is not a detector. It is a threshold low enough to fire on almost everything. The AUC is the honest summary and it is 0.504.

3 Why it failed

Three candidate explanations were tested rather than assumed. The diagnostic script is committed as 07_diagnose_elasticities.py.

3.1 Errors-in-variables — the dominant cause

λ requires differentiating ERA5 boundary-layer-height anomalies. Classical measurement error in a regressor biases the slope toward zero by the reliability ratio. Reverse regression brackets it:

```
forward   d lnC / d lnH         = -0.208    (attenuated)
reverse   1 / (d lnH / d lnC)   = -12.743   (over-stated)
implied reliability ratio         0.016
```

A reliability ratio of **0.016** means the ERA5 BLH anomaly signal is roughly 98% noise once the diurnal cycle is removed. ERA5 derives boundary layer height from a bulk Richardson number and is known to struggle in shallow stable layers — which is the entire Delhi winter night. The elasticity is not small because the physics is weak; it is small because the instrument is.

3.2 The sign is not stable

SPECIFICATION	E1 = DLNC/DLNH	EXPECTED	STATUS
Pooled hourly	-0.043	≈ -1	right sign, tiny
Dense network (>= 8 stations)	+0.061	≈ -1	WRONG SIGN
Daily daytime aggregates	+0.084	≈ -1	WRONG SIGN

THE SYNTHETIC TEST PASSED AND THE REAL DATA DID NOT — WHY

The synthetic generator built the loop in by construction, so the estimator had a clean signal with modest noise and recovered λ to 9%. That validated the ARITHMETIC, not the premise that the signal survives in ERA5. A self-test can only falsify an estimator, never confirm that real data contains what you hope. This is worth saying out loud in the presentation: it is the difference between a team that tested its method and one that believed it.

3.3 A reformulation avoiding PBL height entirely — also fails

If ERA5 BLH is the problem, remove it. Step 8 defines a ventilation index from observed PM2.5 alone:

```
V = ln( PM2.5 pre-dawn  /  PM2.5 afternoon )
```

Large V means the mixed layer grew and cleared the overnight load. V near zero means it did not. It uses only a reference-grade BAM measurement and no derivative of any model field.

- V behaves correctly as a ventilation measure: correlation +0.21 with daytime maximum BLH, +0.15 with mean wind.
- 15% of in-season days show V <= 0 — no net daytime dilution at all.
- **But it does not predict lock-in: AUC 0.554.**

So the failure is not an artefact of ERA5 alone. Something more fundamental is going on, and section 4 identifies it.

4 The finding that redirects the project

If lock-in cannot be predicted from the state before onset, the obvious question is whether it is predictable from anything. It is — from the weather that follows.

PREDICTOR	WINDOW	AUC	VERDICT
Mean wind speed	-24 to 0 h (before onset)	0.434	no skill
Mean wind speed	0 to +48 h (after onset)	0.672	weak
Daytime max BLH	-24 to 0 h (before onset)	0.567	no skill
Daytime max BLH	0 to +48 h (after onset)	0.724	SKILFUL
Ventilation coefficient	-24 to 0 h (before onset)	0.514	no skill
Ventilation coefficient	0 to +48 h (after onset)	0.736	SKILFUL

What this means

Whether a Delhi pollution episode lasts sixteen hours or ten days is decided by the ventilation over the following two days — not by the aerosol–boundary-layer state at the moment it begins. **Lock-in is not a diagnosable regime. It is a forecastable outcome.**

That is a genuine scientific finding, it is measured rather than asserted, and it lands the project exactly where the problem statement wanted it: on coupled forecasting.

The revised positioning

> *We measured whether pollution lock-in can be diagnosed from the present state. It cannot. It is set by the ventilation over the next 48 hours — so we forecast that, and convert it into the time remaining to act.*

This is stronger than the original pitch in three ways. It is backed by a measurement rather than a hypothesis. It explains why a coupled forecast is necessary rather than merely desirable — ventilation is the quantity that decides the outcome, and it depends on the meteorology the PS says must be coupled. And it survives the question that would have destroyed the λ pitch: a reviewer asking what the held-out skill score is.

5 What changes for each module

MODULE	CHANGE
A — Data	Unchanged and validated. The Open-Meteo path removed the CDS dependency entirely. Two new connectors are committed: cpcb_stream.py (authoritative CPCB via data.gov.in, with measured data age) and weather_stream.py, which was 0 bytes and is now a working 72-hour meteorological feed. Priority: close the OpenAQ 2022-2025 gap from CPCB directly so the holdout rests on more than one instrument.
B — Feedback science	The λ mandate is closed. Reassign to the ventilation forecast: define the operational ventilation threshold, quantify how far ahead it can be skilfully forecast, and own the false-alarm cost curve. The diagnostic work in scripts 07 and 08 is the evidence base and should be presented, not hidden.
C — Emulator	Now the centre of the project, with a sharper target. The emulator must forecast the ventilation coefficient and boundary-layer evolution well, because those are the variables shown to determine the outcome. Verification against persistence at 24/48/72 h is still the number that decides the score.
D — Regulatory engine	Unchanged in design, better justified. It now converts a forecast ventilation collapse into lead time and an escalation case. The false-alarm cost threshold matters more than before, because the decision now rests on a forecast rather than an observation.
E — Interface	The primary display becomes the 72-hour ventilation forecast and the intervention window, not a λ gauge. Data age and station count must be visible — 64% of the historical record rests on two or fewer stations and the interface should never imply otherwise.

6 What to do next

ORDER	TASK	OWNER
1	Close the OpenAQ gap: pull CPCB 2022-2025 directly so the holdout rests on a network rather than one monitor.	A
2	Establish how far ahead the ventilation coefficient is skilfully forecastable, using Open-Meteo forecasts verified against the ERA5 archive. This sets the honest lead-time claim.	B
3	Settle the emulator training reference with a written go/no-go on data availability.	C
4	Retune the persistence constants from 3 windows of 3 minutes to a meteorologically meaningful horizon; add the predicted-breach path.	D
5	Rebuild the pitch around the measured finding. The λ slides are withdrawn.	PPT

A negative result obtained in week one, with the evidence to explain it and a measured finding to replace it, is a better position than an unexamined hypothesis carried to the finale.

TEAM DEVENGERS · PS 26082 · FINDINGS REVISION A